Full Stack Assignment – 44

Assignment ID: FSD-44
Title: Threshold-Based Alert Management System
Duration: 3 Hours

1. Objective

Design and develop a **Threshold-Based Alert Management System** that monitors numeric values and generates alerts when defined thresholds are crossed.
The system must enforce threshold rules and ensure **data persistence using a database**.

2. User Roles

OPERATOR

- Register and log in to the system
- Submit monitored values
- View generated alerts related to their submissions

ADMIN

- Register and log in to the system
- Define and manage threshold rules
- View all alerts generated in the system

3. Functional Requirements

Authentication & Authorization

- Implement user registration and login
- Use JWT-based authentication
- Protect all backend APIs using JWT
- Enforce role-based access control on APIs and frontend views

Threshold Configuration

- ADMIN can define threshold rules (example: min/max values)
- Threshold rules must be stored in the database
- Threshold rules must be retrievable dynamically

Value Monitoring

- OPERATOR can submit numeric values
- Submitted values must be stored in the database
- System evaluates submitted values against defined thresholds

Alert Generation

- Alerts must be generated automatically when thresholds are violated
- Alert data must persist in the database
- Alerts must be viewable by authorized users

4. Business Rule (Critical)

A threshold alert must be generated automatically whenever a submitted value exceeds or falls below the defined threshold limits.
Manual alert creation is not allowed.

5. Technical Constraints (Mandatory)

- All threshold, value, and alert data must persist in a database
- Hardcoded or in-memory alert logic is not allowed
- Backend APIs must reflect actual database state
- JWT authentication and role validation are mandatory
- Use environment variables for sensitive configuration

6. Frontend Requirements

- Login and registration pages
- Role-based dashboards for OPERATOR and ADMIN
- Value submission interface for OPERATOR
- Threshold configuration and alert overview for ADMIN
- Alert listing with clear status indicators

UI design is not evaluated; correctness of alert logic is the priority.

7. Submission Requirements (Strict)

GitHub Repository

- Backend source code
- Frontend source code
- Proper folder structure
- Clear commit history

Live Deployment

- Live frontend deployment link
- Live backend deployment link
- Backend must be connected to the database

README.md (Mandatory)

The README file must include:

- Project overview
- Tech stack used

- User roles and permissions
- API endpoints
- Database schema or collections
- Steps to run the project locally
- Live frontend and backend links

8. Evaluation Note

Evaluation focuses on **threshold enforcement, automatic alert generation, database usage, and end-to-end execution**.